

Q1: Simplify and perform the summations below. You may need to use the properties of Kronecker delta and Permutation symbol.

a. $\epsilon_{ijk} \mathbf{A}_i \mathbf{A}_j \mathbf{A}_k :$

$$\epsilon_{ijk} \mathbf{A}_i \mathbf{A}_j \mathbf{A}_k = 0$$

b. $\epsilon_{ijl} \delta_{jl} :$

$$\epsilon_{ijl} \delta_{jl} = 0$$

c. $\mathbf{A}_i \mathbf{B}_j \delta_{ji} - \mathbf{B}_p \mathbf{A}_q \delta_{pq} :$

$$\mathbf{A}_i \mathbf{B}_j \delta_{ji} - \mathbf{B}_p \mathbf{A}_q \delta_{pq} = 0$$

d. $\epsilon_{ijl} \delta_{ip} \delta_{jq} :$

$$\epsilon_{ijl} \delta_{ip} \delta_{jq} = \epsilon_{lpq}$$

e. $\epsilon_{ijl} \delta_{ip} \delta_{jq} \delta_{lr} :$

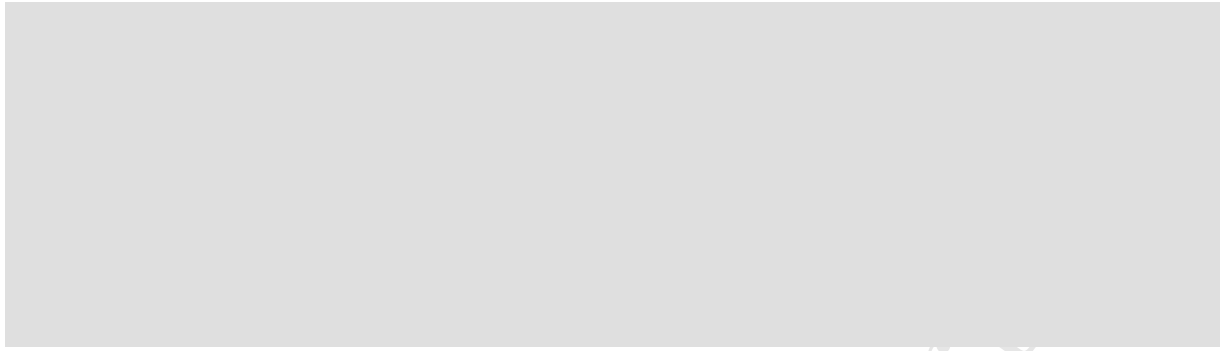
$$\epsilon_{ijl} \delta_{ip} \delta_{jq} \delta_{lr} = \epsilon_{pqr}$$

f. $\delta_{ip} \delta_{pq} \delta_{qj} :$

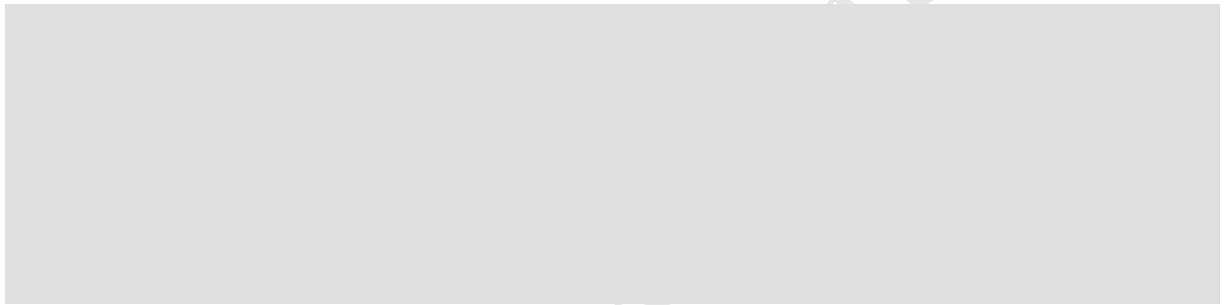
$$\delta_{ip} \delta_{pq} \delta_{qj} = \delta_{ij} \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Q2: Using the index notation verify the vector identities below. (Note that “.” is a dot product)

a. $(\vec{\mathbf{A}} \times \vec{\mathbf{B}}) \cdot (\vec{\mathbf{C}} \times \vec{\mathbf{D}}) = (\vec{\mathbf{A}} \cdot \vec{\mathbf{C}})(\vec{\mathbf{B}} \cdot \vec{\mathbf{D}}) - (\vec{\mathbf{A}} \cdot \vec{\mathbf{D}})(\vec{\mathbf{B}} \cdot \vec{\mathbf{C}}) :$



b. $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{B}(\vec{A} \cdot (\vec{C} \times \vec{D})) - \vec{A}(\vec{B} \cdot (\vec{C} \times \vec{D})) :$



Q3: For matrix $M = [M_{ij}]_{3 \times 3}$ and vector $\vec{V} = [V_i]_{3 \times 1}$, considering $M_{ij} = i + j$ and $V_i = i^2$, calculate the following statements.

$$\vec{M} = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}_{3 \times 3} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}_{3 \times 3}, \quad \vec{V} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}_{3 \times 1}$$

a. $M_{ii} :$

$$\text{[Gray Box]} = 12$$

b. $M_{ij}M_{ij} :$

$$\text{[Gray Box]} = 156$$

c. $V_i V_i :$

$$\text{[Gray Box]} = 98$$

d. $V_i M_{ij} V_j :$

$$= 1008$$

Q4: Let the angles between the primed and unprimed coordinate directions be given by the table shown below. **Determine** the transformation coefficients a_{ij} and **check** the satisfaction of orthogonality conditions.

Hint: For the orthogonality condition one must check the equality

$$a_{ij} a_{ik} = \delta_{jk}.$$

	x_1	x_2	x_3
$\tilde{x}_1$	0°	90°	90°
$\tilde{x}_2$	90°	60°	30°
$\tilde{x}_3$	90°	150°	60°

We have:

$$a_{ij} = \cos(\theta_{ij})$$

Orthogonality condition:

$$a_{ij} a_{ik} = \delta_{jk}$$

a. Number of components:

 = 64

$$\text{---} = 4$$
$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$
$$\varepsilon_{ijl}\varepsilon_{pqr} = \begin{vmatrix} \delta_{ip} & \delta_{iq} & \delta_{ir} \\ \delta_{jp} & \delta_{jq} & \delta_{jr} \\ \delta_{lp} & \delta_{lq} & \delta_{lr} \end{vmatrix}$$